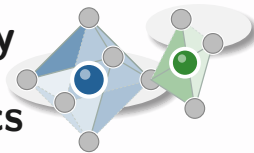


~~Causal~~ – Physical Descriptor Discovery for Interpretable Materials Informatics



*Contracted Observables for Admissible Representation and Stable Interpretation in
Materials Informatics*

Kanchan Sarkar | Axel Groß

Institut für Theoretische Chemie • Universität Ulm

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Dr. Barbara
**Mez-Starck
Stiftung**

Dr. Barbara Mez-Starck Database

MOGADOC

Molecular Gasphase Documentation



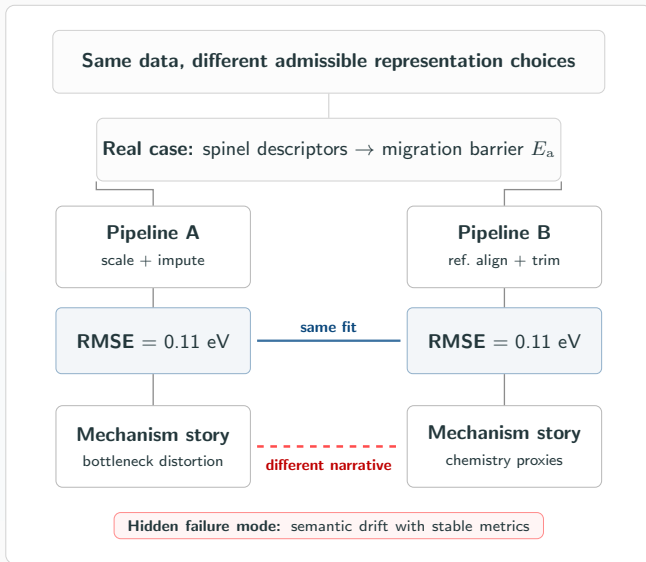
The central question: does good prediction imply physical understanding?

- Two admissible pipelines can give the **same error**.
- They can still imply **different mechanism stories**.
- The failure mode is often *representation semantics*.

Take-home

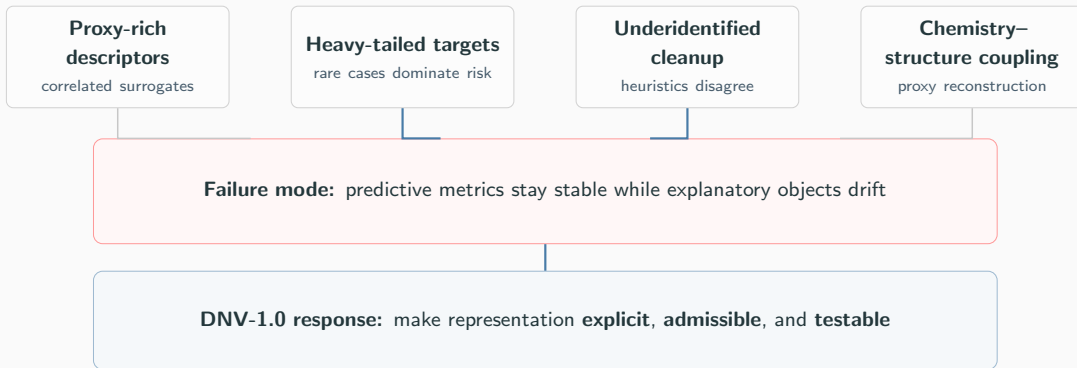
Interpretability is an **invariance claim**.

Stable fit does not imply stable meaning.



Why this problem is acute in materials ML?

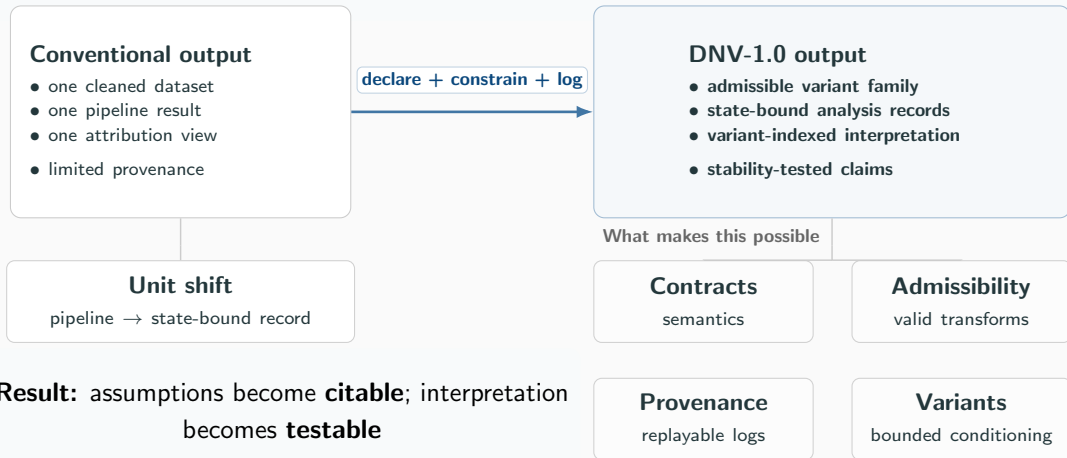
Representation non-identifiability becomes likely when these conditions co-occur



Implication: without governed representation, “important features” and even “the cleaned dataset” may not be uniquely defined scientific objects

DNV-1.0: make representation a governed scientific object

Principle: make representational assumptions explicit and executable, then test whether interpretations persist under admissible change



DNV-1.0 in practice: interface snapshot

Working interface used to run the DNV workflow modules (current build snapshot)



What to notice

- Modular workflow (load, clean, transform, model)
- Data handling and analysis tools in one place
- UI supports reproducible step-by-step processing
- Current build shown; interface can evolve

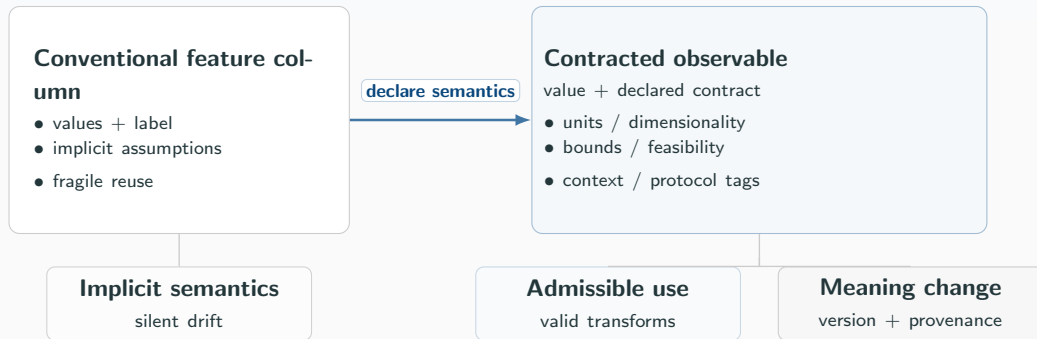
Why show this?

DNV-1.0 is not only a conceptual framework. It is implemented as a practical analysis workflow.

Executable observable contracts (semantics, not just metadata)

A contract turns a feature column into a declared scientific object with executable semantic constraints

e.g. ionic radius, bond length, distortion metric



Operational rule: allow only meaning-preserving transforms.

Otherwise: create a **versioned quantity** with provenance.

We replace one “cleaned table” with an admissible variant family

In heavy-tailed, block-structured data, conditioning is underidentified and should be treated as a set-valued object

Single cleaned table

- one cleanup choice
- hidden conditioning assumptions
- fragile conclusions

replace point
estimate
with admissible set



Admissible variant family

- finite set of dataset states under declared constraints
- replayable variants with bounded edits
- variant-indexed results and interpretations

Admissibility constraints for the variant family

Feasibility gates

mechanism validity
+ contract feasibility

Edit budgets

bounded intervention magnitude
(minimal changes)

Policy constraints

distributional rules
(including tail preservation)

Operational consequence: evaluate interpretation and screening **over the family**, not one cleaned table

Interpretability becomes an invariance test

Promotion rule: a mechanistic interpretation is citable only if it **persists**
across admissible variants and contract-valid proxy substitutions

1. State the claim

record-bound effect / attribution
variant + model state

2. Stress the claim

admissible dataset variants
+ proxy substitutions (same
observables)

3. Decide

Persisting → **stable**
Drifting → **contingent**

Citable unit = analysis record

dataset variant + contract context + method configuration +
model state

Stable fit, drifting meaning

report as **contingent**
(not mechanistic)

Take-home: agreement in predictive error is not evidence of stable meaning

Spinels as a stress test for interpretation

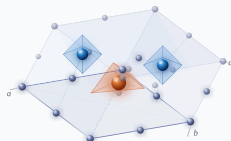
We need a family with comparable framework geometry but changing chemistry

Why spinels are useful for this test

- Common framework: AB_2X_4 spinels
- Chemistry varies while topology stays comparable
- Helps separate structure effects from chemistry proxies
- Target: migration barrier E_a (DFT-computed)



Case-study anchor



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spinel compositions

variant-dependent screened subsets

Mobile ions: Li, Na, Mg, Ca, Zn, Al

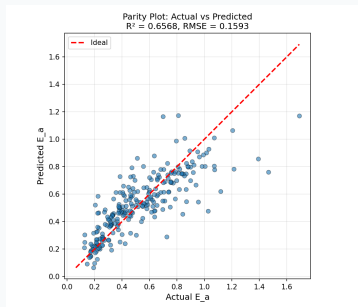
Anions: O, S, Se

Why this is hard: correlated descriptors and chemistry–structure coupling allow proxy shortcuts, so accuracy can remain stable while explanations drift

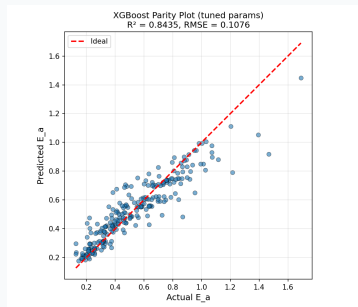
Strong accuracy is achievable, but tail compression remains

Error can be low under the stated protocol, but uncertainty concentrates in the high-barrier tail

Parity plots (same descriptor set; stated split)



LASSO sparse baseline



XGBoost nonlinear model

Interpretation for screening

Models reach ~ 0.1 eV-scale error, but both compress the upper tail. Screening is useful, with elevated uncertainty in the high-barrier regime.

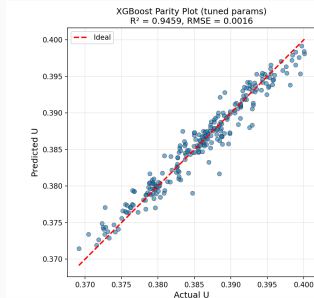
Surrogate audit: predictive success is not yet mechanism

Predictive success can arise from proxy reconstruction rather than the intended mechanism

Audit logic (three-step test)

- **Failure mode:** tabulated-only E_a accuracy may come from proxy reconstruction.
- **Audit task:** predict anion parameter u from ionic radii only.
- **Use:** if this succeeds, E_a accuracy alone is not mechanistic evidence.

Audit example: predict u from radii only



Example outcome $R^2 = 0.95$, $RMSE = 0.0016$
High performance is expected if geometry is recoverable from tabulated size mismatch.

Take-home: the surrogate audit separates “predicts well” from “targets the intended object”

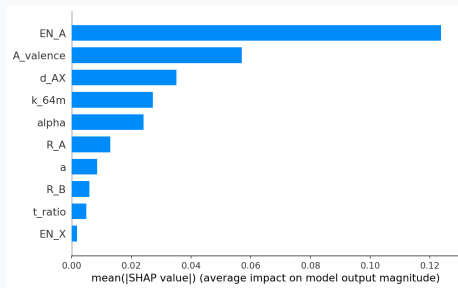
Interpretability result: geometry is the structural channel

Chemistry dominates global variance, while geometry descriptors carry the structural interpretation channel

What the SHAP profile supports

- SHAP is treated as a **state-bound record** (variant-indexed, model state recorded).
- Chemistry controls dominate global variance.
- Geometry terms (d_{AX} , k_{64} , α) carry the structural channel.

XGBoost SHAP summary (variant-indexed model state)



Design-facing output: a variant-indexed shortlist

The screen returns a shortlist with selection stability attached to each entry, not a single pipeline-specific list

Kinetic screening thresholds (dataset-specific, conservative)

- **Li/Na:** $E_a < 0.3$ eV
- **Mg/Ca/Zn/Al:** $E_a < 0.5$ eV

Thermodynamic plausibility filter

$E_{\text{hull}} < 50$ meV is used as a **risk-ranking filter**, not a synthesis guarantee.

Reporting object

Stability labels

family-stable: persists across admissible variants

contingent: appears only under specific conditioning choices

Citable unit

candidate + threshold policy + variant context

candidate $\rightarrow$ **family-stable** / **contingent**

Take-home: selection stability is part of the scientific claim, not a hidden preprocessing side effect

Screening summary (base dataset state)

Applying the kinetic + thermodynamic screen yields strong carrier-dependent attrition, not a uniform reduction across chemistries

Counts by mobile-ion class (base dataset state)					
A-site	z_A	Total	Excellent	Stable	Success rate
Li	1+	40	32	12	30%
Na	1+	45	19	1	2%
Mg	2+	48	31	18	38%
Ca	2+	43	13	5	12%
Zn	2+	36	4	1	3%
Al	3+	37	4	0	0%
Total	—	249	103	37	15%

What to notice

Mg provides the largest stable pool (18 entries), Li shows strong monovalent success (12 stable), and Na exhibits strong kinetic pass rates but severe thermodynamic attrition.

What the shortlist shows (conditional regularities)

The shortlist reveals carrier-dependent chemistry patterns and attrition modes, not a single global screening behavior

Monovalent pattern

Li: highest monovalent success rate; several stable low-barrier chalcogenides.

Na: many kinetically promising entries, but strong thermodynamic attrition.

Design implication

The shortlist is informative because it preserves *which* chemistries survive joint kinetic–thermodynamic screening, not just how many survive.

Multivalent pattern

Mg: largest stable pool; selenides dominate.

Ca: stable candidates are oxides in this dataset/protocol.

Constraint-limited regimes

Zn/Al: strongest constraints under the current chemistry and pathway setup.

Interpretation discipline: these are conditioned regularities (composition set + protocol + admissibility policy), useful as synthesis priors but not universal rules

Interpretability must be treated as an invariance claim

When accuracy can be preserved while semantics drift, mechanistic interpretation is defensible only if it remains stable under admissible representation change

Problem

Representation non-identifiability:
multiple pipelines can achieve
similar error while implying
incompatible physical narratives.

DNV-1.0 move

Contracts + admissibility +
provenance + variant families turn
interpretation into a testable
invariance criterion.

Design-facing output

Stability-labeled interpretations
and variant-indexed shortlists that
are scientifically citable.

Final take-home: DNV-1.0 does not eliminate uncertainty—it makes representational uncertainty explicit, bounded, and auditable

Thank you.



Preprint / group publications